

10-lever prompt-engineering tournament

500 dialogues across 10 prompt levers (n=50 each) · ranked by state accuracy · winner promoted to full n=681

#	Lever	What it does	State acc	(visual)	ROUGE-1	BLEU-4
1	Length budget	Cap tokens per turn	51.96		39.91	12.37
2	Persona	Add teacher persona prefix	51.27		39.60	12.38
3	CoT scaffold	Chain-of-thought prompt	50.89		38.98	9.55
4	Per-state exemplars	1 exemplar per 34-state	50.88		39.42	10.90
5	Negative exemplars	Anti-patterns in prompt	50.71		39.82	10.88
6	Compressed history	Summarize prior turns	47.50		38.79	9.82
7	N-best rerank	Sample 5, pick best	47.33		38.27	10.02
8	Style-matched exemplars	Match exemplars to style	46.72		42.17	12.04
9	Format retry	Retry on schema failure	46.45		39.60	10.55
10	Lexical priors	Topic glossary in prompt	43.89		39.12	9.88

Top-5 levers cluster within 1.3 pp - many help, few dominate. Length-budget winner promoted to full n=681; the locked headline uses it on top of 10-shot stage-balanced exemplars.

ROUGE-1 and BLEU-4 disagree with state accuracy: style-matched exemplars top ROUGE-1 (42.17) while finishing 8th on pedagogy - prefigures the contamination panel.